The Growing-Degree Threshold for Geometric Embeddings of Trees

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Status

This packet records a candidate full answer to Problem 1.6 of Altschuler–Dodos–Tikhomirov–Tyros [1]. The proof combines their random embedding and local-lemma architecture with the Brascamp–Lieb–Luttinger rearrangement inequality [2]. The current verdict is

$$\boxed{\textbf{likely valid full result; human review requested.}}$$

The result determines the threshold up to universal constant factors for the entire range $3 \le \Delta \le N-1$. It does not optimize the constants and does not address the separate optimal-constants Problem 1.5 of [1].

Source problem

A graph $G = (V, E)$ is *geometrically embeddable* into a normed space X if there is a map $\zeta : V \to X$ such that, for all distinct $u, v \in V$,

$$\|\zeta(u) - \zeta(v)\|_X \le 1 \quad \Longleftrightarrow \quad \{u, v\} \in E.$$

The source proves a universal $\Theta\big((\log N)/\log\log N\big)$ threshold when the maximum degree Δ is fixed. On PDF page 3, in Problem 1.6, it asks for the threshold when Δ grows with N.

Problem 1.6 (Trees of growing maximal degree)**.** *Determine an explicit function $m := m(N, \Delta)$ and some universal constants α_0 and α_1 so that the following holds. Any tree on N vertices with maximum degree Δ embeds into any normed space X of dimension at least $\alpha_1 m$. And, no complete tree of degree Δ embeds into any space of dimension less than $\alpha_0 m$.*

Proof overview. Part (A) of Theorem 1.1 is essentially due to Frankl and Maehara. The argument is volumetric: in low dimension there is insufficient volume to pack the required number of disjoint translates of a fixed fraction of the unit ball. An explicit witness is given by the complete rooted $(\Delta-1)$–regular tree of appropriate height.

We now outline the proof of part (B), which is our main contribution. Let $X = (\mathbb{R}^m, \|\cdot\|_X)$

In transcription, Problem 1.6 asks for an explicit function $m(N, \Delta)$ and universal constants α_0, α_1 such that every N-vertex tree of maximum degree Δ embeds into every normed space of dimension at least $\alpha_1 m$, while a complete tree of degree Δ does not embed below $\alpha_0 m$.

Proof Intuition

Put

$$H = \frac{\log N}{\log(\Delta - 1)}.$$

This is the height scale of a complete rooted degree-Δ tree with N vertices. Packing the images of a bipartite independent set in such a tree forces dimension of order

$$q(N, \Delta) = \frac{\log N}{\log(2 + H)}.$$

For the matching upper bound, the source attaches independent uniform vectors from the unit ball of the target norm to the tree edges, then adds a small independent Gaussian regularization. Its proof already works at dimension Aq for every path except the finitely many shortest lengths. The published thin-shell estimate controls such a short path only by $\exp(-m^{\kappa/2})$, which cannot pay for a dependency neighborhood of size $\Delta^{O(1)}$ when Δ is large.

The missing estimate is supplied by rearrangement. Conditional on all but two edge vectors, a short-path failure asks whether the sum of two independent uniform points of an arbitrary symmetric convex body lies in a translate of a slightly enlarged copy of that body. The Brascamp–Lieb–Luttinger inequality says that this probability is maximized by a Euclidean ball. In the Euclidean ball, radii are exponentially concentrated near the boundary, while two independent directions have exponentially small probability of making the fixed obtuse angle required for their sum to have norm near one. Consequently the short-path probability is $O(e^{-cm})$, uniformly over every norm.

This exponential estimate pays for $\Delta^{O(1)}$ in the sparse-dependency regime. If Δ is so large that the local lemma uses its dense N^2 dependency bound instead, then $H = O(1)$ and $q \asymp \log N$, so the same estimate pays for N^2. All remaining path lengths are controlled by the source's $\exp(-cm \log k)$ estimate and its Gaussian completion.

The full theorem

Theorem 1 (Growing-degree threshold)**.** *There are universal constants $A < \infty$ and $N_0 \in \mathbb{N}$ with the following property. Let $N \geq N_0$ and $3 \leq \Delta \leq N - 1$, and define*

$$q(N, \Delta) = \frac{\log N}{\log\left(2 + \dfrac{\log N}{\log(\Delta - 1)}\right)}. \tag{1}$$

Then:

(i) every tree on N vertices with maximum degree at most Δ geometrically embeds into every normed space of dimension at least $\lceil Aq(N, \Delta) \rceil$;

(ii) the breadth-first truncation to N vertices of the complete rooted degree-Δ tree does not geometrically embed into any normed space of dimension less than $q(N, \Delta)/4$.

At the cardinalities of full complete rooted trees, the witness in (ii) is the full complete tree. Thus Problem 1.6 is answered by the explicit threshold function $m(N, \Delta) = q(N, \Delta)$, with universal constants $\alpha_0 = 1/4$ and $\alpha_1 = A$.

Proof

We first prove the norm-uniform estimate that removes the only fixed-degree step from the source proof.

Lemma 2 (Exponential short-path estimate)**.** *Let $\kappa \in (0,1)$ be the absolute exponent in Theorem 2.2 of [1]. There are absolute constants $c, C > 0$ and $m_* \in \mathbb{N}$ such that the following holds. Let $X = (\mathbb{R}^m, \|\cdot\|_X)$ be any normed space with $m \geq m_*$, and let $Y_1, \ldots, Y_k$ be independent random vectors uniformly distributed in its unit ball, where $k \geq 2$. Then*

$$\mathbb{P}\left(\left\|\sum_{i=1}^{k} Y_i\right\|_X \leq 1 + \frac{1}{2m^{1-\kappa}}\right) \leq Ce^{-cm}. \tag{2}$$

The same estimate holds with arbitrary fixed signs in the sum.

Proof. Write $K = B_X$ and

$$t = 1 + \frac{1}{2m^{1-\kappa}}.$$

The body K is centrally symmetric. After increasing m_*, we may assume $t \leq 9/8$.

Condition on $Z = Y_3 + \cdots + Y_k$ (with $Z = 0$ if $k = 2$). For every realization $Z = z$, the conditional probability in (2) is

$$\frac{1}{\mathrm{vol}(K)^2} \int_{\mathbb{R}^m} \int_{\mathbb{R}^m} \mathbf{1}_K(x) \mathbf{1}_K(y) \mathbf{1}_{tK-z}(x+y)\, dx\, dy. \tag{3}$$

Let B be the centered Euclidean ball having the same volume as K. The Riesz rearrangement inequality, equivalently the three-function case of the Brascamp–Lieb–Luttinger inequality [2], bounds (3) by

$$\frac{1}{\mathrm{vol}(B)^2} \int_{\mathbb{R}^m} \int_{\mathbb{R}^m} \mathbf{1}_B(x) \mathbf{1}_B(y) \mathbf{1}_{tB}(x+y)\, dx\, dy.$$

Here the symmetric decreasing rearrangements of $K, K, tK - z$ are respectively B, B, tB. By scaling, the last display is

$$\mathbb{P}\big(\|U + V\|_2 \leq t\big), \tag{4}$$

where U, V are independent and uniform in the Euclidean unit ball.

Write $U = R\Theta$ and $V = S\Phi$, where the directions are independent and uniform on S^{m-1}, independent also of the radii. Since

$$\mathbb{P}(R < 15/16) = \mathbb{P}(S < 15/16) = (15/16)^m,$$

the probability that either radius is smaller than $15/16$ is at most $2(15/16)^m$. On the complementary event, if $\|R\Theta + S\Phi\|_2 \leq 9/8$, then

$$\langle \Theta, \Phi \rangle \leq \frac{(9/8)^2 - R^2 - S^2}{2RS} \leq \frac{(9/8)^2 - 2(15/16)^2}{2(15/16)^2} = -\frac{7}{25}.$$

The middle expression is decreasing in each of $R, S \in [15/16, 1]$.

For a fixed direction, the first coordinate of a uniform point on S^{m-1} has density

$$c_m (1 - s^2)^{(m-3)/2} \mathbf{1}_{[-1,1]}(s), \qquad c_m = \frac{\Gamma(m/2)}{\sqrt{\pi}\, \Gamma((m-1)/2)} \leq \sqrt{m}.$$

It follows, for an absolute $c_0 > 0$, that

$$\mathbb{P}\big(\langle \Theta, \Phi \rangle \le -7/25\big) \le \sqrt{m}\left(1 - \frac{49}{625}\right)^{(m-3)/2} \le C_0 e^{-c_0 m}.$$

Together with the radial estimate, this bounds (4) by Ce^{-cm}, uniformly in z and k. Averaging over Z proves (2). Fixed signs do not alter the distribution because the Y_i are symmetric. □

We now prove Theorem 1. Put

$$D = \log(\Delta - 1), \qquad H = \frac{\log N}{D}, \qquad L = \log(2 + H), \qquad q = \frac{\log N}{L}.$$

Because $3 \le \Delta \le N - 1$, we have $H \ge 1$. Uniformly over this entire range,

$$c\frac{\log N}{\log\log N} \le q \le C \log N. \tag{5}$$

Also, since $H \mapsto \log(2 + H)/H$ is decreasing on $[1, \infty)$,

$$D = \frac{\log N}{H} \le (\log 3) q. \tag{6}$$

Retain the source parameters

$$c_1 = \frac{1}{4}, \qquad c_2 = \min\left\{\frac{1}{4}, \frac{\kappa}{4}\right\}, \qquad \ell_0 = \exp\big((\log N)^{c_2}\big),$$

and its absolute integer $k_0 \ge 3$ from (4.7). Set

$$m = \lceil Aq \rceil, \qquad k_1 = \lfloor H \rfloor,$$

where the absolute constant A will be chosen sufficiently large. From (5), uniformly in Δ,

$$\log \ell_0 = (\log N)^{c_2} = o(q), \qquad k_1 < \ell_0 \tag{7}$$

for all sufficiently large N.

We use the independent uniform edge vectors, path events, dependency graph, and random embedding from Sections 4.1–4.3 of [1]. For a path of length k, let the uniform-vector bad event be

$$\mathcal{L}_{u,v} = \begin{cases} \left\{\left\|\sum_{e \in P(u,v)} \pm Y_e\right\|_X \le 1 + \dfrac{1}{2m^{1-\kappa}}\right\}, & 2 \le k \le k_0, \\ \left\{\left\|\sum_{e \in P(u,v)} \pm Y_e\right\|_X \le k^{c_1}\right\}, & k_0 < k \le \ell_0. \end{cases}$$

The signs depend on the position of the root, but do not change the distribution. Assign every length-k event the local-lemma weight

$$b_k = \begin{cases} \dfrac{1}{2^{k+1} 6 \ell_0 k (\Delta - 1)^k}, & 2 \le k \le k_1, \\ \dfrac{3}{2\pi^2 k^2 N^2}, & k_1 < k \le \ell_0. \end{cases} \tag{8}$$

The first range is allowed to be empty.

For $k \le k_1$, $(\Delta-1)^k \le N$. Since $k \le H = O(\log N)$ and $\ell_0 = N^{o(1)}$, uniformly

$$6k\ell_0(\Delta-1)^k \le N^2 \tag{9}$$

for all sufficiently large N. Claim 4.1 and the calculation in (4.16) of [1] therefore give, for every path-event $\mathcal{L}_{u,v}$,

$$\prod_{\mathcal{L}_{u',v'} \sim \mathcal{L}_{u,v}} (1 - b_{\mathrm{dist}(u',v')}) \ge \frac{1}{2}. \tag{10}$$

It remains to show

$$\mathbb{P}(\mathcal{L}_{u,v}) \le \frac{b_k}{2} \qquad (2 \le k \le \ell_0). \tag{11}$$

We record three elementary comparisons. Their constants depend only on the absolute integer k_0:

$$k_0 < k \le k_1 \implies k\log(2\Delta-2) \le Cq\log k, \tag{12}$$

$$k > k_0 \text{ and } k > k_1 \implies q\log k \ge c\log N, \tag{13}$$

$$2 \le k \le k_0 \text{ and } k > k_1 \implies q \ge c\log N. \tag{14}$$

Indeed, for (12), monotonicity of $x/\log x$ gives

$$\frac{kD}{\log k} \le \frac{HD}{\log H} = \frac{\log N}{\log H} \le Cq,$$

because $H \ge k > k_0$ and $L/\log H$ is then bounded; also $\log(2\Delta-2) \le 2D$. For (13), if $H \ge k_0$, then $k > k_1$ implies $k > H$ and $\log k/L \ge c$; if $H < k_0$, use $k > k_0$ and $L \le \log(k_0+2)$. Finally, (14) follows because $k > k_1$ implies $H < k \le k_0$, so L is bounded.

Suppose first that $2 \le k \le k_0$. Lemma 2 gives

$$\mathbb{P}(\mathcal{L}_{u,v}) \le Ce^{-cm} \le Ce^{-cAq}. \tag{15}$$

If $k \le k_1$, then (6), (7), and (8) give

$$\log\frac{2}{b_k} \le Cq.$$

If $k > k_1$, then (14) gives the same conclusion from $\log(2/b_k) \le 2\log N + o(\log N)$. Thus (11) follows in the short range once A is a sufficiently large absolute constant.

Next assume $k_0 < k \le k_1$. Lemma 3.2 of [1], with $c_1 = 1/4$, gives

$$\mathbb{P}(\mathcal{L}_{u,v}) \le \exp\left(-\frac{m}{8}\log k\right). \tag{16}$$

By (12), (7), and (8),

$$\log\frac{2}{b_k} \le Cq\log k.$$

Choosing $A > 8C$ proves (11) in this range.

Finally assume $k > k_0$ and $k > k_1$. The estimate (16) still holds, while (13) yields

$$\frac{m}{8}\log k \ge cA\log N.$$

On the other hand,

$$\log\frac{2}{b_k} = 2\log N + 2\log k + O(1) = (2+o(1))\log N,$$

because $k \le \ell_0 = N^{o(1)}$. A final universal enlargement of A proves (11). The cases are exhaustive.

Combining (11) with (10), the asymmetric Lovasz local lemma gives positive probability to the event $\mathcal{L}$ on which every nonadjacent pair at tree distance at most ℓ_0 has the required uniform-vector separation.

It remains to check that the Gaussian completion in Section 4.3 of [1] is uniform in Δ. From (5), our dimension satisfies

$$c_A \frac{\log N}{\log\log N} \le m \le C_A \log N. \tag{17}$$

The source uses

$$\delta = \exp\left(-\left(\frac{1}{2} - \frac{c_1}{2}\right)(\log N)^{c_2}\right).$$

For adjacent pairs, for distances $2 \le k \le k_0$, and for distances $k_0 < k \le \ell_0$, its Gaussian upper-tail thresholds contain respectively the factors

$$\frac{\delta^{-1}}{\sqrt{m}\log N}, \qquad \frac{\delta^{-1}}{m^{3/2-\kappa}}, \qquad \frac{\exp((c_1/2)(\log N)^{c_2})}{\sqrt{m}}.$$

The exponential in $(\log N)^{c_2}$ dominates every displayed power of m and $\log N$ uniformly on (17). Equations (4.29)–(4.31) of [1] therefore bound each corresponding conditional bad probability by N^{-2}. For $k > \ell_0$, equations (4.32)–(4.35) give the bound

$$\exp\bigl(-c\,m(\log N)^{c_2}\bigr) \le N^{-2},$$

again uniformly by (17). These estimates use only the Gaussian variables and are unchanged after conditioning on $\mathcal{L}$.

There are fewer than $N^2/2$ unordered vertex pairs. Conditioned on $\mathcal{L}$, the union bound is strictly less than $1/2$, so the random map is a geometric embedding with positive probability. If the ambient normed space has dimension larger than m, carry out the construction in any m-dimensional subspace. This proves part (i).

For part (ii), use the breadth-first truncated complete tree from part (A) of Theorem 1.1 of [1]. It has N vertices, maximum degree at most Δ, a root r, and radius

$$h_0 = \left\lceil \frac{\log\left((N-1)(\Delta-2)/\Delta+1\right)}{\log(\Delta-1)} \right\rceil \le H+1.$$

Suppose it geometrically embeds into a d-dimensional normed space. Since the tree is bipartite, it has an independent set V_0 with $|V_0| \ge N/2$. The radius-$1/2$ balls centered at the images of vertices of V_0 are pairwise disjoint, and all lie in the radius-$(h_0+1/2)$ ball about the image of r. Comparing Lebesgue volumes gives

$$\frac{N}{2} \le (2h_0+1)^d \le (2H+3)^d.$$

For $N \ge 4$, using $2H+3 \le (H+2)^2$, we obtain

$$d \ge \frac{\log(N/2)}{\log(2H+3)} \ge \frac{\log N}{4\log(H+2)} = \frac{q}{4}.$$

Thus no such embedding exists in dimension less than $q/4$. At full complete-tree cardinalities no truncation occurs. This proves part (ii) and the theorem. $\square$

Verification notes

The proof was audited against Sections 3 and 4 of the source. The new input is only Lemma 2; it replaces Lemma 3.3 of the source in the finite range $2 \le k \le k_0$. The event threshold is identical, so no later part of the random embedding changes.

The degree-dependent checks are:

1. the sparse path-dependency inequality (9);
2. the short sparse comparison, paid for by (6);
3. the short dense comparison, where $H = O(1)$ forces $q \asymp \log N$;
4. the transition at $k_1 = \lfloor H \rfloor$, covered by (12)–(14).

All Gaussian estimates are degree-free and use only the uniform dimension window (17). No numerical experiment is used as proof.

The main human-review focus should be the application of rearrangement to the translated third body in (3), and the four exhaustive local-lemma subranges (short/sparse, short/dense, long/sparse, long/dense).

Novelty search and limitations

A bounded literature search on 9 August 2026 checked arXiv:2504.15212, its May 2026 Combinatorica publication, the exact title, the exact phrase "Problem 1.6 (Trees of growing maximal degree)", the authors, and keyword combinations involving geometric embeddings, trees, growing maximum degree, normed spaces, rearrangement, and kissing probability. It found the source paper and the classical rearrangement theorem, but no separate paper claiming to answer Problem 1.6 or giving the threshold (1). Novelty confidence is moderate rather than definitive because the source is recent.

The proof uses a classical theorem in a new place; it does not claim the rearrangement inequality itself as new. The constant A is universal but is not numerically optimized, since the source constant k_0 already depends on implicit absolute constants in the slicing estimates. The result also does not settle the optimal universal constants asked in Problem 1.5.

Human-review recommendation

Review Lemma 2 first. If its translated-set rearrangement step and spherical-cap estimate pass, check the four local-lemma comparisons and the assertion that Section 4.3 of the source is uniform on (17). If those checks pass, this packet gives a complete constant-factor answer to Problem 1.6 for all growing degrees.

References

[1] D. J. Altschuler, P. Dodos, K. Tikhomirov, and K. Tyros, *A universal threshold for geometric embeddings of trees*, Combinatorica 46 (2026), article 20; arXiv:2504.15212v2, `https://arxiv.org/abs/2504.15212`.

[2] H. J. Brascamp, E. H. Lieb, and J. M. Luttinger, *A general rearrangement inequality for multiple integrals*, J. Funct. Anal. 17 (1974), 227–237, `https://doi.org/10.1016/0022-1236(74)90013-5`.